Software Engineering

Lab 4

Vedant Sharma

PES2UG24CS577

Section 5-J

https://github.com/QVedant/SE_Scenario17_Wordle

Files uploaded to github.

Main Bug Prefix

```
Wordle – 5 letters, 6 guesses.
> apple
green gray gray gray gray
> arayy
green green yellow yellow green
> array
green green green green green
Solved!
```

Here “array” is the answer, so the guess “arayy” should have only last “y” as green, and the unnecessary “y” should not be coloured.

```
PS C:\Vedant\PESU\Semester5\SE\Lab\Lab4\17_wordle> python main.py
Wordle – 5 letters, 6 guesses.
> appel
green green green yellow yellow
> allay
green yellow yellow yellow gray
> apple
green green green green green
Solved!
```

Similarly, here “allay” has double “l”, only one of which should have been coloured, yet both are marked yellow.

Task 1 Fix

```
Wordle – 5 letters, 6 guesses.
> appel
yellow yellow gray yellow gray
> graep
green green green yellow yellow
> graap
green green green gray yellow
> ggrape
Enter a valid word of the required length.
> ggrap
green gray yellow yellow yellow
> grape
green green green green green
Solved!
```

Fixed the double/multi occurrence colouring, and the perfect fit match colour hierarchy.

Task 2 Fix

```
PS C:\Vedant\PESU\Semester5\SE\Lab\Lab4\17_wordle> python main.py
Wordle – 5 letters, 6 guesses.
> one
Enter a valid word of the required length.
> gues1
Enter a valid word of the required length.
> guess
gray gray gray gray gray
> secon
gray gray gray gray gray
> third
gray gray gray yellow gray
> fourt
gray gray gray yellow gray
> fifth
gray gray gray gray gray
> sixt
Enter a valid word of the required length.
> sixth
gray gray gray gray gray
Out of guesses!
The word was: array
```

Invalid guess doesn't count for the 6 chances per round. Only 6 valid are counted. It detects incorrect word length.

Task 3 Fix

```
PS C:\Vedant\PESU\Semester5\SE\Lab\Lab4\17_wordle> python main.py
Choose word length (4, 5, or 6): 4
Wordle — 4 letters, 6 guesses.
> apple
Enter a valid word of the required length.
> appl
gray gray gray yellow
> lima
yellow gray gray gray
> ploy
gray green gray gray
> blay
green green gray gray
> blue
green green green green
Solved!
```

4 and 6 length words are a guess option now. Can be set in words.py

Task 4 Fix

```
Solved!

--- Session Summary ---
Outcome: Won
Target word: crane
Valid guesses: 5/6
Total guesses:
1. apple - yellow gray gray gray green
2. falme - gray yellow gray gray green
3. staer - gray gray green yellow yellow
4. grape - gray green green gray green
5. crane - green green green green green
----------------------
PS C:\Vedant\PESU\Semester5\SE\Lab\Lab4\17_wordle>
```

Each guess is saved in a history, with its resulting colour sequence. This helps the user with keeping track.

Chat History

Project materials

17_wordle-main.zip — Wordle game starter project.

Lab_4_VibeCoding_Student_handout.pdf — lab instructions and task requirements.

Gameplay screenshot — used to document the observed bug.

Chronological prompt history

1. Review the project

User prompt: "Please inspect the Wordle project and the Lab 4 handout first. Understand the current code structure and the requirements before making changes."

Purpose: Establish the starter code and lab handout as the reference for the work.

2. Document the gameplay bug

User prompt: "I captured a screenshot of the bug I noticed while playing. Please use it to understand the issue and keep the fix aligned with the existing game."

Purpose: Connect the observed gameplay behavior to the code that needs attention.

3. Task 1 — duplicate-letter feedback

User prompt: "Let's fix Task 1 in feedback.py. Correct the handling of repeated letters while preserving the existing evaluate function, return colors, and overall structure. Why can a simple letter-by-letter check give incorrect feedback when a letter appears more than once?"

Purpose: Fix duplicate-letter scoring while preserving the expected feedback behavior.

4. Task 2 — game flow

User prompt: "Move on to Task 2 in game.py. Make sure the game allows six valid guesses, invalid inputs do not use a turn, a correct guess wins immediately, a loss reveals the target word, and entering q quits."

Purpose: Implement the required game flow and edge-case behavior.

5. Task 3 — word-length modes

User prompt: "Proceed to Task 3. Add selectable four-, five-, and six-letter game modes using main.py and words.py, without breaking the earlier fixes. Why is it useful to keep word selection and game logic separated?"

Purpose: Add configurable word lengths while preserving Tasks 1 and 2.

6. Task 4 — guess history and summary

User prompt: "Now implement Task 4. Show the history of accepted guesses after each turn, and display a final summary for wins, losses, and quits. Include the target word, number of valid guesses, and outcome."

Purpose: Add the requested history and end-of-game summary

Postfix

```
PS C:\Vedant\PESU\Semester5\SE\Lab\Lab4\17_wordle> python main.py
Choose word length (4, 5, or 6): 4
Wordle - 4 letters, 6 guesses.
> gaem
gray gray yellow gray

Guess History:
1. gaem - gray gray yellow gray

> coed
gray gray yellow gray

Guess History:
1. gaem - gray gray yellow gray
2. coed - gray gray yellow gray

> bluu
green green green gray

Guess History:
1. gaem - gray gray yellow gray
2. coed - gray gray yellow gray
3. bluu - green green green gray

> lbbu
yellow yellow gray yellow

Guess History:
1. gaem - gray gray yellow gray
2. coed - gray gray yellow gray
3. bluu - green green green gray
4. lbbu - yellow yellow gray yellow

> lbbb
yellow yellow gray gray

Guess History:
1. gaem - gray gray yellow gray
2. coed - gray gray yellow gray
3. bluu - green green green gray
4. lbbu - yellow yellow gray yellow
5. lbbb - yellow yellow gray gray

> blueeee
Enter a valid 4-letter word.
```

```
> blue
green green green green

Guess History:
1. gaem - gray gray yellow gray
2. coed - gray gray yellow gray
3. bluu - green green green gray
4. lbbu - yellow yellow gray yellow
5. lbbb - yellow yellow gray gray
6. blue - green green green green

Solved!

--- Session Summary ---
Outcome: Won
Target word: blue
Valid guesses: 6/6
Total guesses:
1. gaem - gray gray yellow gray
2. coed - gray gray yellow gray
3. bluu - green green green gray
4. lbbu - yellow yellow gray yellow
5. lbbb - yellow yellow gray gray
6. blue - green green green green
-----------------------
PS C:\Vedant\PESU\Semester5\SE\Lab\Lab4\17_wordle>
```